

SUMEDH GAIKWAD

Pune, Maharashtra

+91-8208700967

SumedhGaikwad.dev@gmail.com

SumedhGaikwad.com

LinkedIn

GitHub

EDUCATION

B.E. Computer Engineering – Honors in AI & ML

Nov 2022 – Jun 2026

Savitribai Phule Pune University (SPPU)

CGPA: 8.83 — SGPA: 9.47 (Latest Semester)

PROJECTS

Adaptive Urban Traffic Signal Optimization Using Reinforcement Learning

GitHub

Final Year BE Project — Python, FastAPI, PyTorch, React, SUMO, TraCI, WebSocket

- Built a D-DQN agent for real-time traffic signal optimization across two network topologies, reducing average waiting time by **58.3%** and increasing vehicle speed by **43.8%** at a single intersection.
- Implemented parallel RL vs. fixed-time simulation with emergency vehicle detection and live monitoring; achieved **12.3% lower waiting time**, leading to SIH 2025 selection and peer-reviewed publication.

Atrio – Real-Time Collaborative Workspace

Live Demo

React, Node.js, Express, MongoDB, Socket.io

- Built a real-time collaborative workspace for shared notes and task management with multi-user authentication and synchronized updates across shared workspaces.
- Implemented Socket.io-based real-time synchronization and deployed the platform for real-world use with a streamlined collaboration workflow.

Finance One – Personal Financial Copilot

Live Demo

TypeScript, Node.js, Express, Prisma, PostgreSQL, TanStack Query, Ollama

- Engineered a personal finance platform around transaction tracking, spending analytics, and flexible weekly, monthly, and quarterly budgeting.
- Designed the backend with JWT authentication, transaction CRUD, budget locking and overlap prevention, and PostgreSQL-backed financial calculations to keep financial logic deterministic and consistent.

WORK EXPERIENCE

Actima Software

Jan 2025 – Aug 2025

Software Engineering Intern

Pune, Maharashtra

- Developed and maintained frontend and backend features across multiple client projects, contributing to REST APIs, database integration, and production-ready web applications.
- Collaborated with senior engineers and clients through Git-based workflows, code reviews, debugging, testing, and iterative delivery to translate business requirements into software solutions.

Virtual2Reality

Jul 2026 – Sep 2026

Freelance Software Engineer

Pune, Maharashtra

- Built a full-stack residential property discovery and advisory platform covering client-facing and administrative workflows for a real estate firm.
- Took the platform from requirements through implementation and deployment, delivering website management and lead management capabilities while working directly with the client.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Java, SQL

Frameworks: React, Node.js, Express, FastAPI, .NET, Vite, TanStack Query

Databases / Storage: PostgreSQL, MongoDB, Prisma, Mongoose, pgvector, S3

AI / ML: PyTorch, Deep RL, DDQN, NLP, LLMs, Chunking, Embeddings

APIs / Real-Time: REST, WebSockets, Socket.io, JWT

Tools: Git, GitHub, Docker, Postman, Linux, Vercel, Render, SUMO, TraCI, Claude Code, Antigravity

AWARDS & CERTIFICATIONS

Smart India Hackathon 2025 – Team VEIG_1

2025

Team Lead

Pune, Maharashtra

- Led the team through multiple screening rounds to college-level selection, developing the Smart Traffic Management System MVP, winning a **Rs. 10,000 cash prize**, and later extending the project into the final-year BE project.

Full Stack Web Development – upGrad

[View Certificate](#)

10-Month Intensive Program

- Hands-on training in MERN, DSA, REST APIs, database integration, and software development.